

1. Risk Analysis (Qualitative) - Probability–Impact Matrix

Exercise 1

A software company identifies five project risks.

Risk	Probability (%)	Impact (Cost Overrun in ₹ Lakhs)
R1	80	10
R2	60	20
R3	30	40
R4	50	15
R5	20	50

Probability Scale:

- Low: <40%
- Medium: 40%-70%
- High: >70%

Impact Scale:

- Low: <₹15 Lakhs
- Medium: ₹15-30 Lakhs
- High: >₹30 Lakhs

Tasks:

1. Categorize each risk.
 2. Plot them on a Probability-Impact Matrix.
 3. Identify which risks require immediate mitigation.
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Exercise 2

For a mobile banking application project:

Risk	Probability	Impact Rating (1-10)
Server Failure	0.75	9
Requirement Change	0.60	7
Key Developer Exit	0.30	8
Cyber Attack	0.20	10
Vendor Delay	0.50	6

Tasks:

1. Classify probability as Low/Medium/High.
2. Classify impact as Low (1-3), Medium (4-7), High (8-10).
3. Create a Probability-Impact Matrix.
4. Rank risks from highest to lowest concern.

2. Monte Carlo Simulation

Monte Carlo Simulation Exercises for Project Time Estimation

Exercise 1: Project Completion Time Estimation

Activity	Optimistic (O)	Most Likely (M)	Pessimistic (P)
Requirements Analysis	5 days	8 days	12 days
Design	10 days	15 days	20 days
Coding	15 days	25 days	40 days

Assume each activity duration follows a triangular distribution.

Tasks

1. Generate 3 simulation trials manually.
2. Compute total project duration for each trial.
3. Calculate average completion time, minimum completion time, and maximum completion time.
4. Estimate the probability that the project finishes within the target duration.

Target duration: 50 days

Exercise 2

A software module development duration is estimated as:

Scenario	Duration (Days)	Probability
Optimistic	15	0.20
Most Likely	20	0.50
Pessimistic	30	0.30

Random numbers generated for 5 simulation runs:

12, 65, 82, 25, 41

Mapping:

- 00-19 → 15 days
- 20-69 → 20 days
- 70-99 → 30 days

Tasks:

1. Determine duration for each run.

2. Calculate average project duration.
3. Estimate probability of completing within 25 days.

Exercise 3

Testing effort for a software project:

Effort (Hours)	Probability
100	0.30
120	0.50
160	0.20

Random numbers:

05, 31, 72, 88, 15

Tasks:

1. Simulate 5 outcomes.
2. Calculate expected testing effort.
3. Find percentage of simulations exceeding 120 hours.

3. Decision Tree Analysis

Exercise 1

A company can invest in automated testing.

- Cost of automation = ₹2,00,000

If successful:

- Gain = ₹10,00,000
- Probability = 0.70

If unsuccessful:

- Gain = ₹1,00,000
- Probability = 0.30

Tasks:

1. Draw decision tree.
2. Calculate EMV (Expected Monetary Value).
3. Determine whether automation should be adopted.

Exercise 2

Two alternatives are available.

Option A: In-house Development

- Success probability = 0.80

- Profit = ₹15,00,000
- Failure probability = 0.20
- Loss = ₹4,00,000

Option B: Outsourcing

- Success probability = 0.60
- Profit = ₹12,00,000
- Failure probability = 0.40
- Loss = ₹1,50,000

Tasks:

1. Construct decision tree.
2. Compute EMV of each option.
3. Recommend the better alternative.

4. Sensitivity Analysis

Exercise 1

A project estimates effort using:

$$Cost = Team\ Size \times Duration \times ₹50,000$$

Baseline:

- Team Size = 8
- Duration = 10 months

Suppose duration varies by $\pm 20\%$.

Tasks:

1. Calculate project cost for 8, 10, and 12 months.
2. Determine percentage change in total cost.
3. Identify sensitivity to project duration.

Exercise 2

Software development budget:

Parameter	Baseline
Developers	12
Salary/Developer/Month	₹80,000
Duration	9 Months

Variables may vary:

- Developers $\pm 25\%$

- Duration ±15%
- Salary ±10%

Tasks:

1. Calculate revised project costs.
2. Determine which parameter causes maximum cost variation.
3. Rank parameters based on sensitivity.

5. Failure Mode and Effects Analysis (FMEA)

Exercise 1

For a software deployment project:

Failure Mode	Severity (S)	Occurrence (O)	Detection (D)
Configuration Error	8	4	5
Database Failure	10	3	4
API Integration Failure	7	5	6
Requirement Omission	9	6	7

Formula:

$$RPN = S \times O \times D$$

Tasks:

1. Calculate RPN for all failures.
2. Rank failure modes.
3. Identify top two issues requiring action.

Exercise 2

A cloud migration project reports:

Failure Mode	S	O	D
Data Loss	10	2	4
Security Misconfiguration	9	4	6
Downtime	8	5	3
Backup Failure	9	3	5

Tasks:

1. Compute RPN.
2. Rank risks.
3. Suggest which failure should be addressed first.

6. Fault Tree Analysis (FTA)

Exercise 1

Top Event: **Software System Crash**

Caused by:

- Database Failure (0.05)
- Server Failure (0.03)

Connected through an OR gate.

Tasks:

1. Calculate probability of system crash.
2. Determine contribution of each basic event.
3. Identify most critical component.

Exercise 2

Top Event: **Release Delay**

Occurs when BOTH events happen:

- Requirement Change (0.30)
- Vendor Delay (0.20)

Connected through an AND gate.

Tasks:

1. Calculate probability of release delay.
2. If Vendor Delay increases to 0.40, recalculate.
3. Determine percentage increase in delay risk.

7. SWOT Analysis (Numerical) for Software Project Risk Management

Exercise 1

A software firm assigns weights (1-10) and scores (1-5).

Strengths

Factor	Weight	Score
Experienced Team	9	5
Strong QA Team	8	4

Weaknesses

Factor	Weight	Score
Limited Budget	8	2
Legacy Technology	7	2

Opportunities

Factor	Weight	Score
New Market Demand	9	5

AI Integration	7	4
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Threats

Factor	Weight	Score
New Competitor	8	3
Security Threats	9	2

Formula:

$$\text{Weighted Score} = \text{Weight} \times \text{Score}$$

Tasks:

1. Compute weighted scores.
2. Total each SWOT category.
3. Identify strongest strategic area.

Exercise 2

A project manager evaluates risk strategy.

SWOT Factor	Weight	Score
Skilled Developers	10	5
Requirement Volatility	8	2
Cloud Adoption Opportunity	9	4
Regulatory Changes	7	2

Tasks:

1. Calculate weighted score for each factor.
2. Compute overall project strategic score.
3. Determine whether strengths and opportunities outweigh weaknesses and threats.

This exercise helps students use **SWOT Analysis for Risk Identification and Risk Assessment** in Software Project Management.

Exercise 3:

A software company is developing an **Online Banking Application** that allows customers to perform fund transfers, bill payments, and account management through web and mobile platforms.

SWOT Analysis

Strengths (S)

- Experienced software development team
- Strong support from bank management
- Proven software development methodology

Weaknesses (W)

- Limited cybersecurity expertise

- Tight project schedule
- Dependence on third-party payment gateway

Opportunities (O)

- Increasing adoption of digital banking
- Availability of cloud-based security services
- Government initiatives promoting cashless transactions

Threats (T)

- Cyberattacks and data breaches
- Frequent changes in banking regulations
- Competition from fintech companies

Tasks

1. Identify Risks

Using the SWOT factors, identify **five project risks**.

2. Risk Assessment

Assign:

- Probability (1-5)
- Impact (1-5)

Calculate:

$$Risk\ Score = Probability \times Impact$$

3. Risk Response Planning

Suggest a suitable response strategy:

- Avoid
- Mitigate
- Transfer
- Accept

4. Prioritize Risks

Rank the risks according to their risk scores.

Solution

Step 1: Risk Identification

Risk ID	Risk Description	Source
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R1	Security vulnerabilities due to limited cybersecurity expertise	Weakness
R2	Project delay because of tight schedule	Weakness
R3	Payment transactions fail due to third-party gateway issues	Weakness
R4	Data breach caused by hacking attempts	Threat
R5	Requirement changes due to new banking regulations	Threat

Step 2: Risk Assessment

Risk ID	Probability (1-5)	Impact (1-5)	Risk Score
R1	4	5	20
R2	4	4	16
R3	3	4	12
R4	5	5	25
R5	3	5	15

Step 3: Risk Response Planning

Risk ID	Response Strategy	Action
R1	Mitigate	Hire cybersecurity consultants and conduct security training
R2	Mitigate	Add schedule buffer and monitor milestones regularly
R3	Transfer	Establish SLA with payment gateway provider
R4	Mitigate	Implement penetration testing and multi-factor authentication
R5	Accept/Mitigate	Keep contingency budget and regulatory monitoring team

Step 4: Prioritization

Rank	Risk ID	Risk Score
1	R4	25
2	R1	20
3	R2	16
4	R5	15
5	R3	12

Inference: The highest-priority risk is **R4 (Data Breach)** with a risk score of **25**, requiring immediate mitigation efforts.